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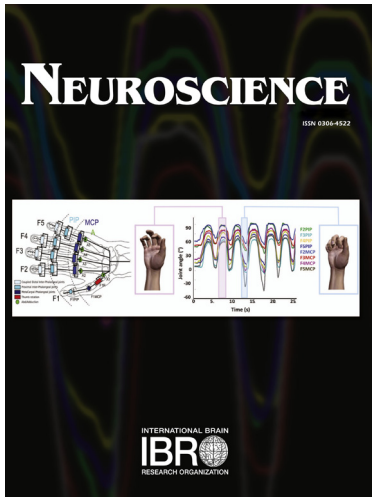

Review

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State-Dependent Alpha Peak Frequency Shifts: Experimental Evidence, Potential Mechanisms and Functional Implications

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Abstract

Neural populations produce complex oscillatory patterns thought to implement brain function. The dominant rhythm in the healthy adult human brain is formed by alpha oscillations with a typical power peak most commonly found between 9 and 11 Hz. This alpha peak frequency has been repeatedly discussed as a highly heritable and stable neurophysiological "trait" marker reflecting anatomical properties of the brain, and individuals' general cognitive capacity. However, growing evidence suggests that the alpha peak frequency is highly volatile at shorter time scales, dependent on the individuals' "state". Based on the converging experimental and theoretical results from numerous recent studies, here we propose that alpha frequency variability forms the basis of an adaptive mechanism mirroring the activation level of neural populations which has important functional implications. We here integrate experimental and computational perspectives to shed new light on the potential role played by shifts in alpha peak frequency and discuss resulting implications. We further propose a potential mechanism by which alpha oscillations are regulated in a noisy network of spiking neurons in presence of delayed feedback.

Keywords: Alpha peak frequency, activation state, frequency tuning, network oscillations, information gating, neural computation

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Introduction

Alpha activity is one of the most salient dynamical features of both the resting and awake human brain. These pervasive electrical oscillations, most usually found within the 8-12 Hz frequency band, have emerged as one of the most robust predictors of brain state. Since their discovery by Hans Berger in 1929 (Berger, 1929), alpha oscillations have been the center of numerous neuroscientific investigations. One of the most consistent observations about alpha oscillations is their sustained variability. Indeed, the frequency of human alpha oscillations have been found to change according to age, cognitive state, sensorimotor processing, sleep and across a wide range of neurological disorders (Uhlhaas and Singer, 2006). To which extent this variability reflects shifts in neural processing states and/or timely transitions in neural sampling strategies is currently poorly understood. A better understanding of the mechanisms involved in shaping the frequency of endogenous alpha oscillations would represent a significant progress in linking macroscopic brain recordings to the dynamics of cortical networks and the underlying computational schemes.

In light of converging experimental and computational results from numerous recent studies, we here wish to link different perspectives into a unified framework. We propose that alpha peak frequency variability does not represent random fluctuations but rather it forms the basis of an adaptive mechanism mirroring the activation level of neural populations. By adaptation, we mean that the frequency of synchronous neural activity is self-regulated and is tuned autonomously, adjusting to task demands to regulate input and information processing. We discuss one well-studied candidate mechanism supporting such frequency regulation in a neuron network model with delayed feedback, suggesting that this newly identified mechanism might be a generic property of cortical networks.

Over the last few years, numerous studies have raised the fascinating prospect of using non-invasive stimulation to manipulate endogenous alpha oscillations in order to alter cognition and alleviate symptoms of neurological disorders. Given that cortical systems possess the intrinsic ability to regulate their peak frequency, we believe it is critical to better understand what these frequency transitions mean in terms of neural dynamics and information processing. By

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exploring the mechanisms involved, we might be able to devise more accurate control stimulation paradigm and better understand the role of neural oscillations in brain function.

State-dependent alpha frequency transitions in humans

It is well documented in the literature that the resting-state *alpha peak frequency* (APF) can differ remarkably between and within individuals. Specifically, in close analogy to brain volume and general cognitive development, the APF increases from early childhood until adolescence, remains stable during adulthood and then starts to decrease with age and/or manifestation of neurological disorders such as Alzheimer's disease (cf. Klimesch, 1999). Nevertheless, even age-matched healthy individuals exhibit variations in APF with a mean standard deviation of about 1 Hz (reviewed in Klimesch, 1996, 1997). Other neurological conditions associated with slowing of the APF include reduced oxygen supply to the brain such as in cerebral ischemia (Van der Worp et al., 1991) and in carotid artery occlusion (Mosmans et al., 1983) as well as burnout syndrome (van Luijtelaar et al., 2010), cancer-related fatigue (Zimmer et al., 2015) and neurogenic/neuropathic pain (Boord et al., 2008; Sarnthein et al., 2006). However, the APF increases again following carotid endarterectomy (Uclés et al., 1997; Vriens et al., 2000) as well as after neurosurgical pain relief therapy (Sarnthein et al., 2006). The resting state APF is highly heritable (Posthuma et al., 2001; van Beijsterveldt and van Baal, 2002), and it has been shown to qualify as a stable neurophysiological trait marker in healthy younger and older adults (Grandy et al., 2013).

However, growing experimental evidence indicates that substantial APF fluctuations exist at much smaller time scales (e.g. one second), where such fluctuations were found positively correlated with regional cerebral blood flow (rCBF) in a brain network comprising areas that are associated with the modulation of attention and preparedness for external input (Jann et al., 2010). Moreover, a number of studies reported shifts of the APF as a result of an acute intervention across various conditions. For example, Angelakis et al. (2004) observed significant increases in APF immediately after the Digit Span working memory task however, only for those participants whose APF was lower than the sample median before the task (Angelakis et al.,

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2004). More recently, Haegens et al. (2014) as well as Maurer et al. (2015) reported a load-dependent acceleration of the APF during performance of a N-back and a modified Sternberg task, respectively (Haegens et al., 2014; Maurer et al., 2015). In addition, Gutmann et al. (2015) found a significant increase of the resting state APF following a strenuous bout of cycling exercise whereas following moderate intensity exercise, the change of APF was substantially smaller, and not significant (Gutmann et al., 2015). This indicates the extent of physiological changes induced by physical exercise, and thus, most probably, the input to the brain, is associated with the extent of APF acceleration. In a similar manner, Hülsdünker et al. (2016) reported increases in APF during more demanding balance task performance (e.g. unipedal vs. bipedal balancing) (Hülsdünker et al., 2015). Moreover, in this study a significant positive correlation was found between APF and the amount of postural sway probably suggesting that the APF increases with increased allocation of attentional resources as well as increased somatosensory input to the brain as a consequence of postural instability. The APF was also found to increase during tonic noxious thermal pain stimulation (Nir et al., 2010) as well as after acute administration of nicotine (Lindgren et al., 1999), caffeine (Bchir et al., 2006; Tiffin et al., 1995) and buflomedil - a vasoactive drug (Saletu et al., 1984). However, and this is important in view of the aforementioned computational predictions, the change of APF is bidirectional depending on the individual's state. Specifically, reductions of the APF were observed during the ideational reproduction of sorrow and fear emotions whereas emotions of joy and anger induced a shift of the APF in the opposite direction (Kostyunina and Kulikov, 1996). Furthermore, significant deceleration of the APF was reported during Qi Gong (Zhang et al., 1988) and meditation state (Saggar et al., 2012). Moreover, the APF was positively correlated with the Coma-Recovery Scale - Revised (CRS-R) score in patients suffering from disorders of consciousness. That is, the lower the total CRS-R score and thus, the expected degree of consciousness, the lower the APF (Lechinger et al., 2013). Interestingly, the APF shifts reported in coma patients are substantially different from those reported during sleep (Lechinger et al., 2015) or anesthesia (cf. Purdon et al., 2015) despite the fact that the reduced degree of consciousness is common for all these situations.

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The cumulative pattern of experimental results clearly indicates that task/state-dependent changes in APF is a widespread phenomenon not limited to specific types of tasks. Moreover, a number of studies suggest the APF changes as a function of task demands or input intensity. This, as well as the general concept of the proposed APF modulation mechanism is illustrated in Figure 1 and 2, respectively. In a similar way as heart rate, intraindividual shifts may be interpreted in terms of task/state-dependent adaptations whereas interindividual differences may reflect differences in biology, and with respect to alpha, differences in brain connectivity (Valdés-Hernández et al., 2010).

Although the above described research lends strong support to our hypothesis stating that alpha frequency variability forms the basis of an adaptive mechanism mirroring the activation level of neural populations, some important methodological aspects should be considered. First, a real frequency shift cannot – at least if the APF is measured from the power spectrum – be distinguished from an asymmetric power desynchronization favoring the lower or the upper alpha band (for an illustration cf. Figure 1 in Klimesch et al., 1996). Empirical evidence suggests that intraindividual APF shifts may indeed be due to asymmetric desynchronization because intraindividual task-dependent APF shifts are often – in contrast to interindividual APF variances – not related to differences in cognitive performance (Klimesch et al., 1996; Maurer et al., 2015). However, an important aspect thereby is how fast intraindividual instantaneous frequency shifts occur. That is, when APF is measured from the spectrum the time range is in seconds (due to the time/frequency resolution issue). But instantaneous shifts may occur in the range of a single alpha period. Such “frequency sliding” can be computed via the temporal derivative of the phase angle time series resulting in a time series of instantaneous frequencies (Cohen, 2014). Importantly, the speed of the instantaneous APF, as computed by this method, has been shown to predict the temporal resolution of visual perception (Samaha and Postle, 2015) suggesting the APF can indeed be a limiting factor for behavioral outcomes, at least with regard to perception (see below). In addition to these aspects, so far the vast majority of studies limited their analyses to changes in APF at posterior sites where the peak is best detectable. However, some of those studies that analyzed APF shifts across different cortical regions found significant region-specific effects (e.g. Klimesch et al., 1993; Nir et al., 2010) or at

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least a clear tendency towards it (Haegens et al., 2014). Hence, although this needs further investigation, consistent with our hypothesis the alpha frequency modulation may differ between regions depending on the cortical networks engaged/disengaged in a particular situation.

Mechanisms of alpha frequency regulation: a computational approach

Numerous modeling studies have uncovered mechanisms by which neural systems regulate their collective synchronous firing during sensory processing. For instance, gamma activity, a high-frequency (30-150 Hz) oscillation produced by local synaptic interactions between excitatory pyramidal cells and inhibitory interneuron, is known to fluctuate within a wide range of frequencies during visual stimulation, such as during the presentation of spatially periodic gratings (Buszaki and Wang, 2012). Gamma frequency shifts commonly reflect changes in stimulation statistics (Ray and Maunsell, 2010) and are supported by circuit motifs in which inputs shape the timing of reciprocal interactions between pools of excitatory pyramidal cells and inhibitory interneurons (Wang and Buzsáki, 1996; Whittington et al., 1995; Wilson and Cowan, 1972). The frequency of resulting driven neural oscillations typically scales with input magnitude (Brunel and Wang, 2003; Ermentrout and Kopell, 1998; Jadi and Sejnowski, 2014). All these represent examples in which oscillatory neural activity adapts to context and sensory input statistics. However, the mechanism by which slower oscillations, such as the alpha rhythm, also experience frequency shifts has remained poorly understood.

Alpha activity is believed to rely on larger scale processes (Hindriks et al., 2014) supported by delayed network interactions (Cabral et al., 2014), and to build on slower and more global inhibitory processes (Klimesch et al., 2007; Womelsdorf et al., 2014). Recent computational work has provided new perspectives on how alpha rhythmic activity might be tuned by afferent and feedback inputs in presence of delayed and non-linear connectivity in large-scale brain network models (Lefebvre et al., 2015). In these studies, the authors have shown that non-linear neural networks exhibiting synchronous alpha oscillations in presence of delayed feedback adapt their frequency and power according to input statistics of both feedforward and feedback stimuli.

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To better illustrate the mechanism at work, let us consider a generic large-scale network of interacting neural populations. One may write the dynamics of recurrently connected sub-networks as (Lefebvre et al., 2015, Hutt et. al., 2016)

$$s^{-1}\frac{d}{dt}\boldsymbol{u} = -\boldsymbol{u} + \boldsymbol{W}\boldsymbol{f}[\boldsymbol{u}(t-\tau)] + \boldsymbol{I} \qquad (1)$$

Where u refers to the mean somatic membrane potential of a population of neurons, W is a matrix of synaptic weights, τ is a mean axonal propagation delay and I corresponds to a task-related input (e.g. noise) that scales with task demands and/or the intensity of sensory drive. The dynamics evolve on a time scale set by the choice of the time constant s. The neuron response function f maps the magnitude of neural activity to its output firing rate. This function typically displays a sigmoidal shape: it increases gradually with u but plateaus afterwards as firing activity saturates (Deco et al., 2008). Equation (1) above generically exhibits slow non-linear oscillations: neural populations develop synchronous dynamics over a wide range of parameter values, whose frequency is mainly determined by the delay and time scale chosen. Mean field analysis of Eq. (1) above has revealed that such systems implement an on-line gain control mechanism in which oscillatory activity induced by the large scale delayed loops adapts the intensity of the input I. causing the emergent synchronous dynamics of the system to accelerate with respect to some baseline input I_o. An example of this mechanism in a network model is plotted in Figure 3.

The relationship between the frequency f_α of endogenous alpha oscillations and the input intensity I can be approximated to first order by the equation

$$f_\alpha \approx f_\alpha^o(1 + \Delta[I - I_o]) \qquad (2)$$

in which f_α^o is the baseline alpha frequency, I_o the baseline input level and $\Delta[I - I_o] \approx -\frac{2}{\sqrt{\pi}\bar{w}}(I - I_o) + O(I^2)$ is a frequency shift that scales with the input difference $I - I_o$ and a mean synaptic weight $\bar{w}$. Equation 2 above states that whenever the input I increases with respect to I_o, the network frequency accelerates. The opposite is also possible: if the mean input I decreases with respect to I_o, so does the frequency. More specifically, changes in APF are linked to fluctuations in input statistics, such as through a change in mean and/or variance

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(Lefebvre et al. 2015). Linear stability analysis performed on Eq. (1) has further revealed that such frequency shifts are bounded: acceleration occurs until oscillations becomes unstable whenever $I \gg I_o$, being then replaced by asynchronous activity (Hutt et al., 2016). Contingent on parameter values considered in these studies, frequency shifts were shown to occur within a physiological range for the alpha frequency i.e. between 8 Hz and 15 Hz. A similar range of frequencies was also observed in a highly detailed cortical network model experiencing non-linear acceleration in presence of pulsatile stimulation (Herrmann et al., 2016).

Many studies have pointed out that dynamic task-dependent inputs cause a change in the response function of neural populations, becoming increasingly linear under the action of noise (Gammaitoni, 1995; Sutherland et al., 2009; Tuckwell et al., 2009). In presence of network oscillations, such as alpha activity, this linearization is reflected in the dynamics of the neurons by a change in frequency. This implies that the frequency of alpha activity generated by synchronous neural population mirrors the firing rate of its constituent neurons: during tasks, behavior and/or arousal, involved neural populations will rise their spiking with respect to baseline. These increased firing rates will engage non-linear feedback loops, effectively changing the system's response function and the specifics of its emergent oscillations. In contrast, during rest and states of quietness, the spiking activity decreases, which is also reflected by a decrease in the frequency of alpha activity. The effect discussed above has been replicated in a variety of computational models, notably using electromagnetic stimulation, where high frequency inputs have been found to emulate a state of high arousal and thus provoke a change in the alpha peak frequency (Herrmann et al., 2016). As a whole, these computational findings collectively suggest that activation-dependent frequency transitions within the alpha band constitute a generic property of non-linear cortical networks. Considered in conjunction with the postulated role of synchronous neural activity in information processing, the computational results summarized above suggest that input-induced frequency shifts represent a mechanism by which neural systems adapt features of endogenous rhythms to context and stimuli.

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On the functional role of alpha frequency shifts

We suggest the above reviewed state-dependent fluctuations of the APF are due to shifts of a distinct (individual) alpha frequency and are not due to the modulation of a population of different alphas as traditional work has suggested (e.g., Basar and Bullock, 1992), and they are functionally relevant. This view is also supported by numerous findings showing that alpha phase plays an important functional role for perception (Romei et al., 2012; Samaha et al., 2015; for review see e.g. Hanslmayr et al., 2011). The point here is that phase is meaningful only for a distinct frequency and not for a compound waveform stemming from a population of different frequencies. Due to their different frequencies, a population of alphas would sum (at the scalp/magnetic field measurement level of the EEG or MEG outside the brain) to a compound waveform, whose 'phase' has no longer a physiological meaning because peaks and troughs are more or less random superpositions of different oscillations. More recent studies go even a step further by showing that the alpha frequency can be externally triggered (entrained) by non-invasive brain stimulation, and the entrained alpha frequency predicts the temporal resolution of the individual's visual perception in a phase-dependent manner. For example, applying transcranial alternating current stimulation (tACS) to the parieto-occipital cortex and concurrently recording EEG, (Helfrich et al., 2014) could demonstrate that 10 Hz tACS increases parieto-occipital alpha activity and synchronizes cortical oscillators with similar intrinsic frequencies to the entrainment frequency. This suggests the entrainment of oscillations within the resonance frequency can phase align the internal oscillator to the external oscillator. Importantly, tACS modulated individuals' target detection performance in a visual oddball task in a phase-dependent fashion. This highlights the causal role of alpha oscillations for visual perception. Furthermore, (Cecere et al., 2015) found that individual APF correlates with the temporal window in which two auditory beeps paired with a single flash induce the double flash illusion such that the slower the alpha frequency, the larger the temporal window of illusion. A similar mechanism of sensory sampling was later proposed by Samaha and Postle (2015). Importantly, Cecere et al. (2015) used occipital tACS to modulate the individual APF towards slower or faster alpha frequency. This manipulation respectively enlarged or shrunk the temporal window of illusion, suggesting the APF causally supports the

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temporal integration of information across the senses. A possible mechanism of action of this novel "entrainment" approach introduced by Cecere et al. (2015) is proposed by Romei et al. (2016) (Figure 3).

From a computational perspective, state-dependent adaptation of alpha oscillations could play many parallel functional roles. As many experimental studies have pointed out, alpha rhythms appear to be involved in information processing through periodic cycles of sampling and sensory integration, whose pace correlates strongly with perceptual performance and behavior (see above). As such, the state-dependent acceleration/deceleration of the APF implements a temporal sampling strategy, where the sampling rate is adjusted as a function of task demands to increase performance and optimize temporal integration and representation of sensory information. In addition to this adaptive sampling scheme, the flexibility of state-dependent frequency control could also play an important role in routing the flow of information across brain areas. Indeed, according to the communication through coherence framework (Fries, 2005, 2015), distant brain areas establish communication through frequency- and phase-specific synchronization. To engage different targets, brain networks need to tune the frequency of their synchronous oscillations as a function of context to route task-related information towards the relevant areas. In this framework, it would prove insightful to use brain stimulation to entrain local alpha activity and see how information propagation, and thus functional communication, is modulated by state-dependent changes in APF.

Conclusion and suggestions for future research

Based on the reviewed evidence we propose that different alpha frequencies do not reflect a population of different alphas. We assume that they are due to frequency shifts which are suggestive of a coding scheme in which accelerated alpha oscillations during cognitive load and/or overt behavior constitutes an optimal dynamic state for sensory sampling, neural computation and inter-area communication. As such, the activation-dependent, self-regulation mechanism of alpha peak frequency adds to a growing body of evidence indicating that the alpha rhythm may dictate the resolution at which information is sampled and/or processed by

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cortical neurons (Cecere et al., 2015; Samaha et al., 2015; Samaha and Postle, 2015). From the perspective of the communication through coherence hypothesis (Fries, 2005, 2015) in which brain communication builds upon stable, coherent synchrony between different neural populations, local brain networks need to tune their endogenous frequency to engage a variety of phase relationships with distal functional networks, routing task-related information towards relevant neural targets. Understanding the source of these dynamic frequency shifts will hopefully help us better understand the involvement of alpha oscillations in neural coding, information gating, and access the more general role of dynamic rhythms in the healthy and diseased brain.

While the relationship between task difficulty and APF shifts appears to be consistent across a wide range of studies, the role of peak alpha power is less clear. Computational work suggests that APF acceleration is commonly accompanied of a concomitant decrease in peak power (Hutt et al., 2016), although the opposite can be seen if networks are brought in a different excitability state prior to input onset (Lefebvre et al., 2015). We note that changes in peak power are also contingent on the nature of the input correlations driving the neural populations involved. This question warrants further investigation.

From the perspective of the here proposed framework we would like to suggest several ideas and hypotheses to be studied in future experimental and computational research. Previous research has emphasized the importance of alpha reactivity for behavior. For example, good as compared to bad memory performers show significantly more power in the upper alpha band during resting-state as well as a large decrease in alpha power (desynchronization) during actual task performance (cf. Klimesch, 1999). However, these findings refer to alpha power, and the link between alpha frequency reactivity and behavior remains comparatively poorly studied. The here proposed framework suggests that task-related APF shifts are functionally relevant and thus, it is predicted that APF reactivity should be most pronounced in task relevant brain areas, and it should be related to behavioral outcome measures. The two studies cited above (Klimesch, 1996 and Maurer et al., 2015) that failed to find a significant relationship between the task-related shifts of the APF (i.e. APF reactivity) and differences in task performance challenge this prediction but the evidence is not sufficient to draw any final

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conclusions. In fact, a recent study by Hülsdünker et al. (2015) suggests that a shift of the APF may support maintaining upright posture as the balancing task becomes more demanding. Importantly, in this study, the topographical pattern of alpha power changes found with increasing balance task demands did not correspond to that of the APF shift thus, indicating the APF shift is not simply a "byproduct" of asymmetric power synchronization/desynchronization. Therefore, further studies are needed that tackle this issue using not only cognitive but also perceptual and/or sensorimotor tasks. These experiments should include a change/progression of task difficulty to provoke APF shifts, and both, alpha phase as well as instantaneous alpha frequency should be analyzed and related to behavior. The amplitude of at least alpha oscillations should also be determined and the lower and upper alpha bands should be analysed separately. This will allow examination of the rather complex relationship between alpha amplitude and frequency. Furthermore, computational methods such as independent component analysis could be helpful to identify different cortical alpha sources with distinct peaks (Onton and Makeig, 2005). Moreover, work on intracranial alpha oscillations, and particularly work on cycle per cycle fluctuations, is needed to conclusively verify whether frequency shifts stem from task relevant sources/networks. Especially, local field potential recordings from different sources in non-human primates (e.g. Haegens et al., 2015) and invasive recordings in human patients (e.g., Crone et al., 1998) would be very helpful in that regard. With respect to the above discussed tACS research, the here proposed framework would assume that a locally restricted tACS-induced shift of the APF should facilitate primarily those perceptual and/or cognitive processes that are within the main functions of the stimulated brain region, and that are closest associated with the function of alpha (e.g. access to memory, sensory processing).

To assess potential causal relationships between behavior and state-dependent APF fluctuations within this new framework, one might also use the approaches detailed in numerous theoretical and experimental studies exploring neural population coding in presence of global oscillations (Chalk et al., 2016) and coupling between alpha activity and higher frequencies (Jensen et al., 2016), notably using strategies that rely on sequential processing of sensory information (Jensen et al., 2014) and the selective functional inhibition of distractor

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networks (Jensen and Mazaheri, 2010). Finally, combining such experiments with non-invasive brain stimulation in order to interfere with task-related APF shifts, and the behavioral consequences of this, seems to be a very promising approach to shed additional light on the functional role of task-related APF shifts.

As mentioned, recent computational studies suggest that APF scales with the excitability of neural circuits and reflects the changing statistics of sub-cortical and cortico-cortical inputs. As such, task and state-dependent fluctuations in APF could be used to infer the dynamics of pre-synaptic inputs and/or to estimate correlations between those and the activity of other cortical areas at various time-scales, opening up new perspectives on spatio-temporal interactions and oscillatory control mechanisms in the brain. In addition, such correlations may further provide new viewpoints on network interactions underlying the characteristic spectral transitions observed within the alpha band during altered brain states such as across sleep stages and anesthesia (e.g. see Purdon et al., 2015).

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Conflicts of interest

No relevant conflicts of interest.

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Figure legends

Figure 1: A selection of studies demonstrating alpha peak frequency accelerations as a function of task demands or pain stimulus intensity. Alpha peak frequency increases from rest to task performance with low tasks demands to task performance with high task demands for working memory tasks (Haegens *et al*., 2014, *Neuroimage* and Maurer *et al*., 2015, *Brain Topogr*, not shown) as well as during sensorimotor control (Hülsdünker *et al*., 2016, *Front Hum Neurosci*). A similar pattern of alpha frequency acceleration occurs during application of tonic heat stimuli where noxious heat stimuli (10.3 Hz) lead to greater alpha peak frequency increase when compared to innocuous stimuli (Nir *et al*., 2010, *Brain Res*). These results suggest the shift in alpha frequency is (i) a widespread phenomenon not limited to specific types of tasks, and (ii) it depends on the level of cortical involvement.

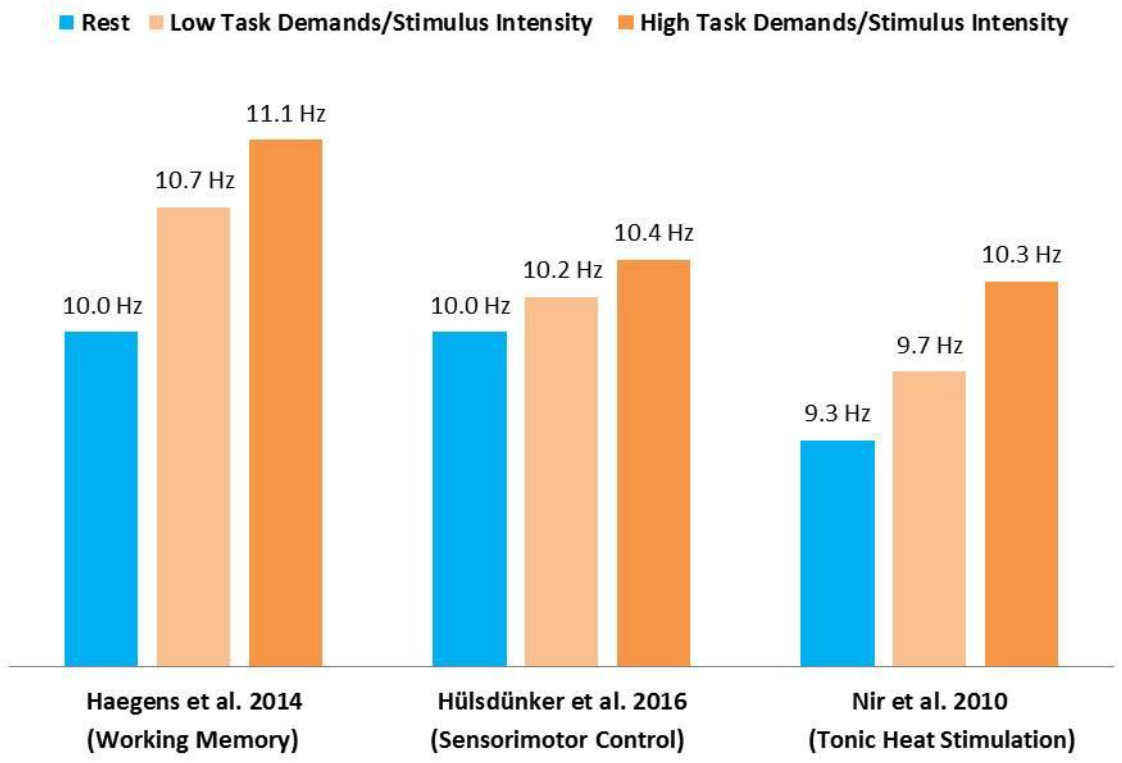

Figure 2: Hypothetical modulation of alpha peak frequency as a function of age and activation/deactivation level. Alpha peak frequency, which starts to decrease with age after reaching a maximum during adolescence, can fluctuate substantially on shorter time scales. The direction of alpha frequency modulation depends on the level of activation or deactivation of distinct neural populations. Cortical activation by means of increased information processing demands, selective emotional states or analeptic drugs have been associated with a speed up of alpha frequency (orange). In contrast, cortical deactivation as a result of relaxation, some

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meditation sates or coma has been shown to slow down the frequency of alpha oscillations (blue).

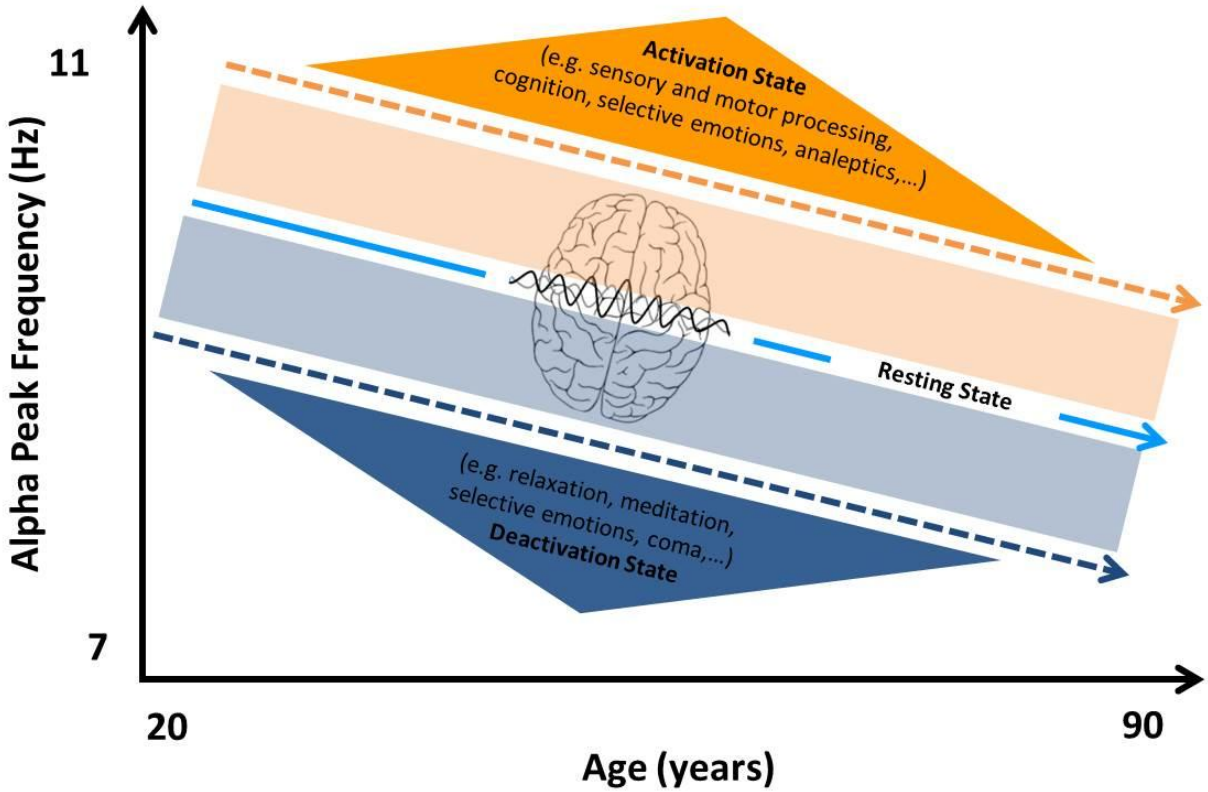

Figure 3: Numerical simulations showing an alpha peak shift in a noisy network of spiking neurons in presence of delayed feedback. (A) In the rest state, baseline afferent input (grey arrow) translates into weak-intensity feedback interactions with other intermediate networks, causing local cortical networks built of excitatory (triangle, E) and inhibitory (circle, I) cells to stabilize into slow oscillatory alpha activity. Here, afferent inputs were modelled using Gaussian white noise and was assumed to drive excitatory cells only. (B) In contrast, when inputs are increased with respect to baseline, strong recruitment of non-linear feedback causes an acceleration of synchronous activity toward higher frequencies. (C) Power spectral densities associated to rest (blue) and task (orange) states. One can see the shift from ~10 Hz to ~11 Hz under the action of increased afferent inputs alone, without significant change in peak power. (D) Alpha peak frequency of the network model versus afferent input intensity. As task demands increase, neuronal spiking also increases, resulting in a gradual acceleration of endogenous alpha oscillations. Model and parameters used for simulations are fully detailed in Herrmann *et al.,* 2016.

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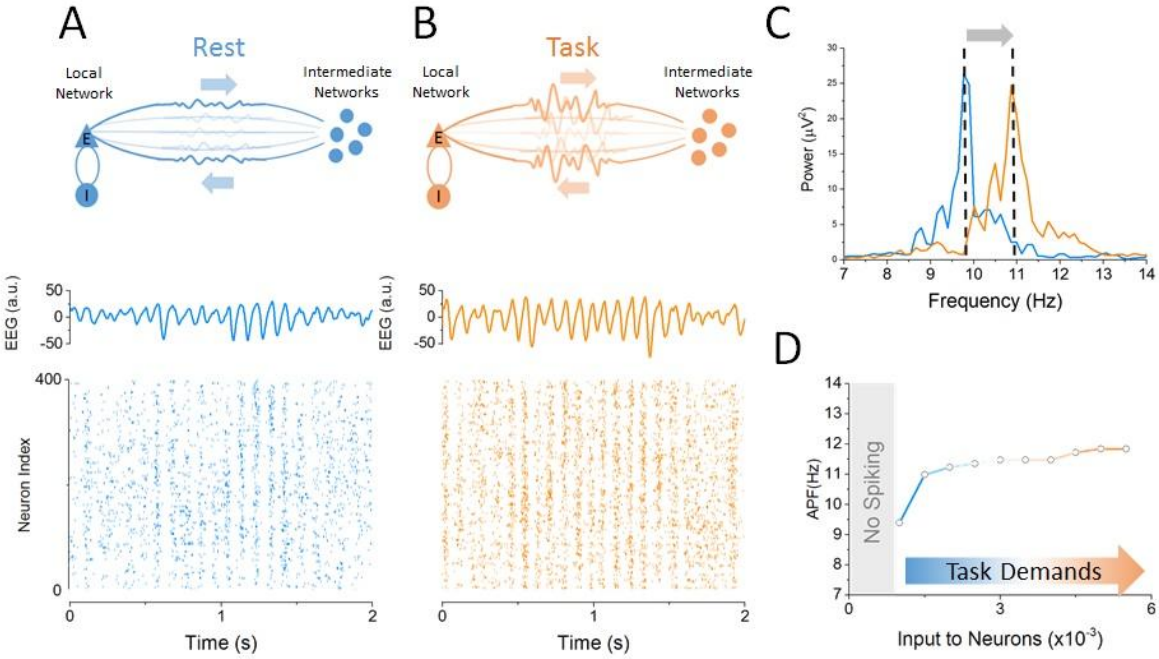

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Highlights

- Alpha oscillations have emerged as one of the most robust predictors of brain state
- Alpha frequency fluctuates at shorter time scales as function of behavioral demands
- Adaptation of alpha peak frequency is hypothesized to influence perception and access to memory
- Cortical connectivity implements a gain control mechanism regulating alpha frequency
- Alpha frequency transitions can be externally triggered (entrained) by non-invasive brain stimulation

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